**Subject:** ACS 547, Noise Control Applications - Module 2 Assignment

**Date:** March 24, 2025 (Submitted)

---

# Problem 1 - Washing Machine Mount Design (1 DOF)

## Problem 1a

The Matlab code for this problem is listed in Appendix 1.

Figure 1 shows the load force produced by the 1 kg imbalanced mass at a 0.5 m radius from the center of the washing machine as a function of angular speed in RPM.

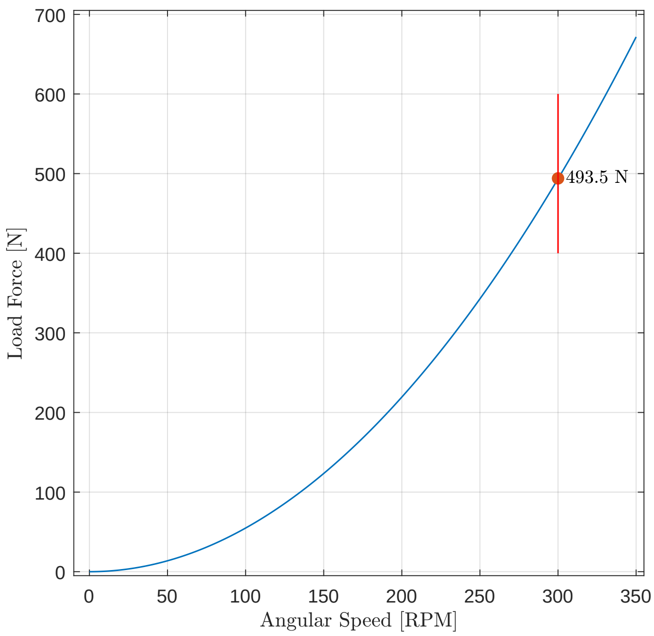


Figure 1: Load force of 1 kg imbalanced mass at a 0.5 m radius versus washing machine angular speed.

The force amplitude is 493.5 N at a frequency of 5 Hz when the washing machine operates at 300 RPM.

## Problem 1b

The static displacement of the load in the washing machine must be less than 1 cm (i.e., the deflection is calculated with only the 10 kg load mass). The corresponding mount stiffness, $k_s$, is **9,810** $\frac{N}{m}$.

The natural frequency, $w_o$, of the mount is **1.5 Hz** (or 9.4 $\frac{\text{radians}}{s}$).

## Problem 1c

Figure 2 shows the transmission loss of the system for a set of damping ratios ranging from 0.001 to 1.

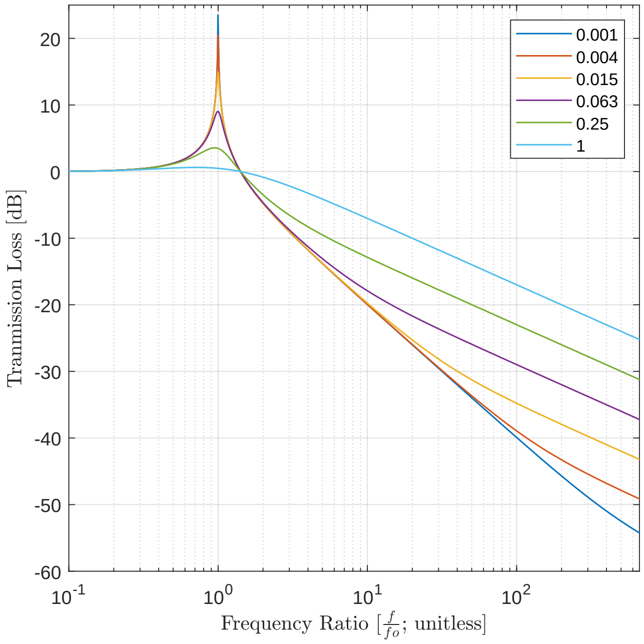


Figure 2: Transmission loss of the system. Its natural frequency is 1.5 Hz.

## Problem 1d

Figure 3 shows the force applied to the foundation as a function of RPM.

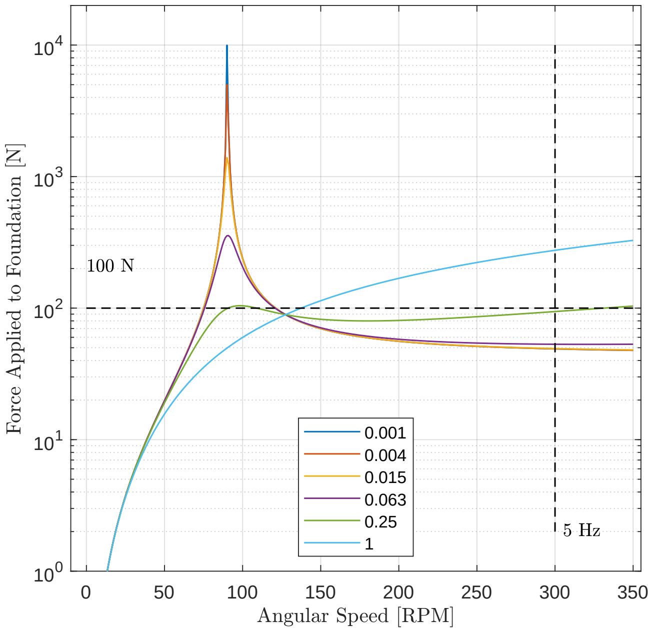


Figure 3: Force applied to the foundation versus RPM.

There is no damping ratio for which the applied force is less than 100 N across the whole operating range. The best damping ratio is $\zeta = 0.25$. With this damping ratio, the force applied to the foundation still exceeds the 100 N target in two regions: 91 to 109 RPM and 331 to 350 RPM. In both regions the force is about 104 N.

## Problem 1e

Figure 4 shows the admittance as a function of RPM.

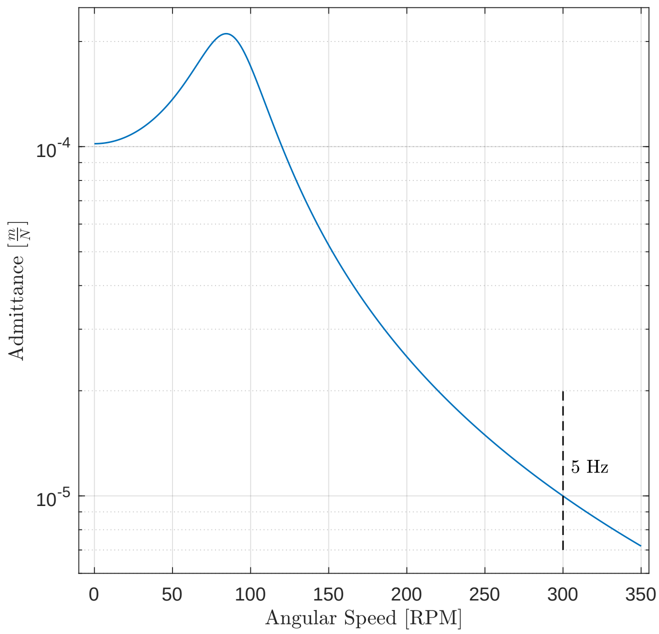


Figure 4: Admittance versus RPM.

The admittance at 300 RPM is 1e-5 $\frac{\mathrm{m}}{\mathrm{N}}$.

## Problem 1f

Figure 5 shows the displacement of the washing machine as a function of RPM.

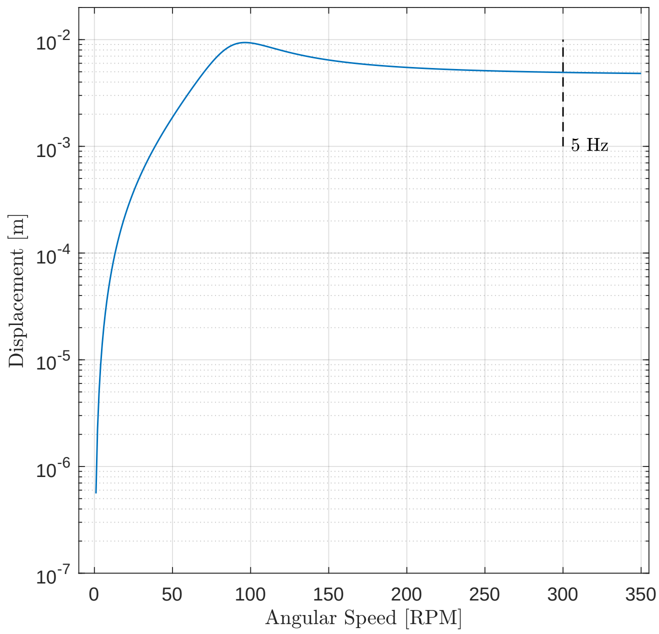


Figure 5: Displacement versus RPM.

The displacement of the washing machine at 300 RPM is 5 mm. This amount of displacement for the 300 RPM operating state seems excessive. The washing machine undergoes larger momentary displacements as is spins up to the 300 RPM operating state, in particular around 100 RPM.

# Problem 2 - Washing Machine Mount Design (2 DOF)

## Problem 2a

The Matlab code for this problem is listed in Appendix 2.

Figure 6 illustrates the two-stage vibration mount considered here (see slide 8 of 42 for Lecture 14 on Monday, March 3, 2025). Modeling will be done using discrete viscous damping elements.

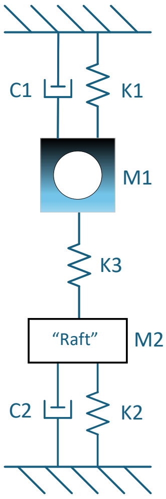


Figure 6: Two-stage vibration mount system for the washing machine.

## Problem 2b

The static displacement of the load in the washing machine must be less than 1 cm (i.e., the deflection is calculated with only the 10 kg load mass). The corresponding mount stiffness, $k_s$, is **9,810** $\frac{N}{m}$.

The natural frequency, $w_o$, of the mount is **1.5 Hz** (or 9.4 $\frac{\text{radians}}{s}$).

Table 1 lists the design parameters for the design enclosure.

| **Wall** | **Width [m]** | **Height [m]** | **Area [$m^2$]** | **Aspect Ratio** | **Correction** | **Compliance [$\frac{m^3 \cdot s^2}{kg}$]** |
|---|---|---|---|---|---|---|
| Top | 1.25 | 1.5 | 1.875 | 1.2 | 1.8 | 1.42e-6 |
| Side 1 | 1.5 | 3 | 4.5 | 2.0 | 0.6 | 2.57e-6 |
| Side 2 | 1.5 | 3 | 4.5 | 2.0 | 0.6 | 2.57e-6 |
| Side 3 | 1.25 | 3 | 3.75 | 2.4 | 0.5 | 1.49e-6 |
| Side 4 | 1.25 | 3 | 3.75 | 2.4 | 0.5 | 1.49e-6 |

Table 1: Enclosure design summary.

The estimated insertion loss without the hole is 13.5 dB, which exceeds the target loss of 13.

The estimated insertion loss with the hole is 1.2 dB, based on the hole depth correction of Deng (1998). The hole reduces the required insertion loss. The intake silencer needs to provide 11.8 dB of insertion loss.

# 1 Appendix - Matlab Code for Problem 1

## 2 Appendix - Matlab Code for Problem 2

## 3 Appendix - Matlab Code for Problem 3

## 4 Appendix - Matlab Code for Problem 4